CALCULUS · UNIT 4: APPLICATIONS OF DERIVATIVES

Package Design Studio

Use optimization, related rates, and curve analysis to design packaging for a juice company.

Honors

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Fresh Squeeze Juice is launching a new drink and needs packaging that uses as little material as possible. The bottling plant fills cone-shaped tanks, and the marketing team wants to understand the sales curve.

You will use derivatives to design the package, control the filling process, and analyze sales.



Can Volume

355 cm³

(12 fl oz)

Box Sheet

30 cm by 20 cm

cardboard

Learning Goals

- I can model and solve optimization problems with constraints and costs.
- I can solve related-rates problems with changing geometry.
- I can use derivatives, theorems, and linearization to justify conclusions.

Project Timeline

When	What you will do
Day 1	Cost Optimization
Day 2	Tank Dynamics
Day 3	Sales Theorems
Final	Design Proposal + reflection

Part 1: Cost Optimization

Day 1 · A 355 cm^3 can. The top and bottom are thicker and cost 0.05¢ per cm^2; the side costs 0.02¢ per cm^2.

Design Data Cards

Open Box	Can	Tank
Cut squares of side x	V = 355 cm^3	Inverted cone
from a 30 × 20 sheet	minimize surface area	height 6 m, radius 2 m

Fill Rate	Sales	Cost
2 m^3 per minute	$S(t) = -t^3 + 12t^2 + 60t$	Aluminum 0.02¢ per cm^2
	thousand cans, t months	

1. Write the cost C(r).

2. Find the radius that minimizes cost and the height.

3. Compare with the minimum-area can. Why is the cost-optimal can taller?

4. Find the minimum cost per can and the savings per million cans versus r = 3.84 cm.

Part 2: Tank Dynamics

Day 2 · Inverted cone h = 6 m, r = 2 m. Fills at 2 m^3/min and leaks at 0.1h m^3/min.

5. Write $\frac{dV}{dt}$ and then $\frac{dh}{dt}$ as a function of h.

6. Find $\frac{dh}{dt}$ when h = 3 m.

7. Is there a height where the level stops rising? Is it inside the tank?

8. Use linearization of V(h) = $\frac{\pi h^3}{27}$ at h = 3 to estimate the volume added when h goes from 3 to 3.1 m.

Part 3: Sales Theorems

Day 3 · $S(t) = -t^3 + 12t^2 + 60t$, $0 \le t \le 12$.

9. Use the EVT to find the absolute max and min of S on [0, 12].

10. Find the maximum of the growth rate S' on [0, 12] and justify it.

11. Write the design proposal: can dimensions, tank controls, and the sales timeline, each justified with calculus.

Final Product: Design Proposal

Your final product must include:

- Cost-optimized can with comparison
- Tank model with leak and linearization
- EVT and growth-rate analysis
- Savings estimate
- Design proposal

Reflection

R1. How did changing the cost structure change the optimal shape?

R2. Why does linearization work well for small changes?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Optimization, related rates, and curve analysis are set up, solved, and justified correctly. Goes beyond the minimum with a creative, realistic choice.	Optimization, related rates, and curve analysis are set up, solved, and justified correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes diagrams, derivative work, justifications, and recommendations. Polished, organized, and visually engaging.	Proposal includes diagrams, derivative work, justifications, and recommendations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.